Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Sql Views

2520030222

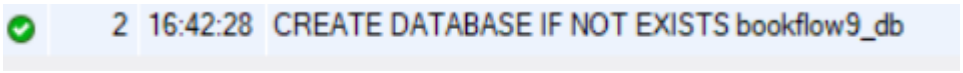
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Section – 3

Create database:

```
CREATE DATABASE IF NOT EXISTS bookflow9_db;
```

Output:



Use database:

```
USE bookflow9_db;
```

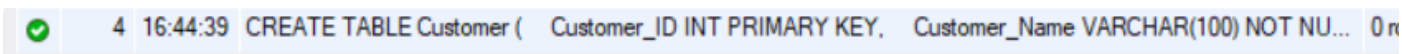
Output:



Create customer table:

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
```

Output:



Create account table:

```sql
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output:

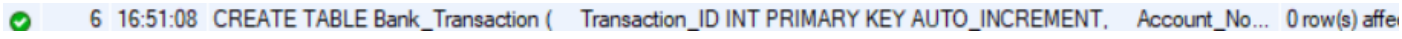


Create Bank Transaction Table:

```sql
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output:

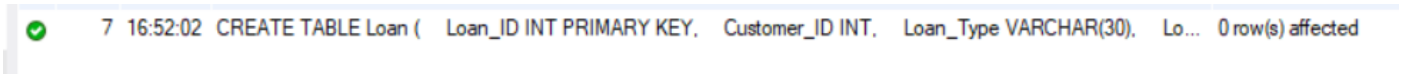


Create Loan Table:

```sql
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

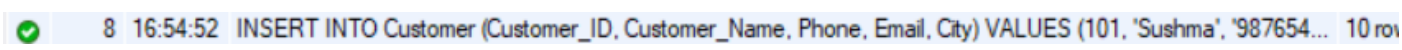
Output:



Insert Customer Data:

```sql
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Sushma', '9876543210','ravi@gmail.com', 'Hyderabad'),
(102, 'Sowmya', '9876543211','priya@gmail.com', 'Vijayawada'),
(103, 'Mokshagna', '9876543212','arjun@gmail.com', 'Bangalore'),
(104, 'Akhila', '9876543213','sneha@gmail.com', 'Chennai'),
(105, 'Likhitha', '9876543214','kiran@gmail.com', 'Hyderabad'),
(106, 'Aswitha', '9876543215','anil@gmail.com', 'Delhi'),
(107, 'Vaishnavi', '9876543216','meena@gmail.com', 'Mumbai'),
(108, 'Vignesh', '9876543217','rahul@gmail.com', 'Pune'),
(109, 'Shanmukh', '9876543218','lakshmi@gmail.com', 'Hyderabad'),
(110, 'Jumpa', '9876543219','suresh@gmail.com', 'Vijayawada');
```

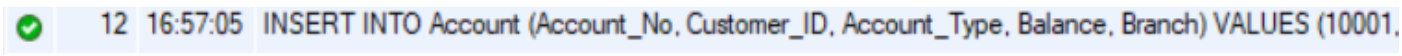
Output:



Insert Account Data:

```
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad'),
(10006, 106, 'Savings', 65000, 'Delhi'),
(10007, 107, 'Current', 150000, 'Mumbai'),
(10008, 108, 'Savings', 35000, 'Pune'),
(10009, 109, 'Savings', 85000, 'Hyderabad'),
(10010, 110, 'Current', 110000, 'Vijayawada');
```

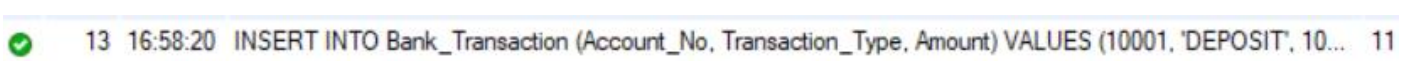
Output:



Insert Transaction Data:

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10001, 'WITHDRAW', 5000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000),
(10006, 'DEPOSIT', 12000),
(10007, 'DEPOSIT', 25000),
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);
```

Output:



Insert Loan Data;

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10001, 'WITHDRAW', 5000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000),
(10006, 'DEPOSIT', 12000),
(10007, 'DEPOSIT', 25000),
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);
```

Output:

```
13  16:58:20  INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...  11 row(s)
```

Insert loan data:

```
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type,
 Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),
(502, 102, 'Education Loan', 1000000, 6.5),
(503, 103, 'Car Loan', 800000, 8.2),
(504, 104, 'Personal Loan', 500000, 10.5),
(505, 105, 'Home Loan', 4000000, 7.2),
(506, 106, 'Car Loan', 900000, 8.5),
(507, 107, 'Business Loan', 3000000, 9.0),
(508, 109, 'Personal Loan', 600000, 10.0);
```

Output:

```
14  16:59:55  INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type,  Loan_Amount, Interest_Rate) VALUES (501, 101, '...  8 row(s) affected Records: 8  Dupli
```

Display Original Tables:

```
SELECT * FROM Customer;
```

Output:

Customer_ID	Customer_Name	Phone	Email	City
101	Sushma	9876543210	ravi@gmail.com	Hyderabad
102	Sowmya	9876543211	priya@gmail.com	Vijayawada
103	Mokshagna	9876543212	arjun@gmail.com	
104	Akhila	9876543213	sneha@gmail.com	Chennai
105	Likhitha	9876543214	kiran@gmail.com	Hyderabad
106	Aswitha	9876543215	anil@gmail.com	Delhi
107	Vaishnavi	9876543216	meena@gmail.com	Mumbai
108	Vignesh	9876543217	rahul@gmail.com	Pune
109	Shanmukh	9876543218	lakshmi@gmail.com	Hyderabad
110	Jumpa	9876543219	suresh@gmail.com	Vijayawada
NULL	NULL	NULL	NULL	NULL

arjun@gmail.com

```
SELECT * FROM Account;
```

Output:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Bank_Transaction;
```

Output:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 16:58:20
2	10001	WITHDRAW	5000.00	2026-08-15 16:58:20
3	10002	DEPOSIT	15000.00	2026-08-15 16:58:20
4	10003	WITHDRAW	20000.00	2026-08-15 16:58:20
5	10004	DEPOSIT	5000.00	2026-08-15 16:58:20
6	10005	WITHDRAW	10000.00	2026-08-15 16:58:20
7	10006	DEPOSIT	12000.00	2026-08-15 16:58:20
8	10007	DEPOSIT	25000.00	2026-08-15 16:58:20
9	10008	WITHDRAW	5000.00	2026-08-15 16:58:20
10	10009	DEPOSIT	20000.00	2026-08-15 16:58:20
11	10010	WITHDRAW	15000.00	2026-08-15 16:58:20
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Loan;
```

Output:

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00
NULL	NULL	NULL	NULL	NULL

Create A View For All Customers:

```
CREATE VIEW Customer_View AS
SELECT *
FROM Customer;
SELECT * FROM Customer_View;
```

Output:

Customer_ID	Customer_Name	Phone	Email	City
101	Sushma	9876543210	ravi@gmail.com	Hyderabad
102	Sowmya	9876543211	priya@gmail.com	Vijayawada
103	Mokshagna	9876543212	arjun@gmail.com	Bangalore
104	Akhila	9876543213	sneha@gmail.com	Chennai
105	Likhitha	9876543214	kiran@gmail.com	Hyderabad
106	Aswitha	9876543215	anil@gmail.com	Delhi
107	Vaishnavi	9876543216	meena@gmail.com	Mumbai
108	Vignesh	9876543217	rahul@gmail.com	Pune
109	Shanmukh	9876543218	lakshmi@gmail.com	Hyderabad
110	Jumpa	9876543219	suresh@gmail.com	Vijayawada

Create A View For Customer Basic Information:

```
CREATE VIEW Customer_Basic_View AS
SELECT
    Customer_ID,
    Customer_Name,
    City
FROM Customer;
SELECT * FROM Customer_Basic_View;
```

Output:

Customer_ID	Customer_Name	City
101	Sushma	Hyderabad
102	Sowmya	Vijayawada
103	Mokshagna	Bangalore
104	Akhila	Chennai
105	Likhitha	Hyderabad
106	Aswitha	Delhi
107	Vaishnavi	Mumbai
108	Vignesh	Pune
109	Shanmukh	Hyderabad
110	Jumpa	Vijayawada

Create A View For Account Information:

```
CREATE VIEW Account_View AS
SELECT
    Account_No,
    Account_Type,
    Balance,
    Branch
FROM Account;
SELECT * FROM Account_View;
```

Output:

Account_No	Account_Type	Balance	Branch
10001	Savings	50000.00	Hyderabad
10002	Savings	75000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	45000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	65000.00	Delhi
10007	Current	150000.00	Mumbai
10008	Savings	35000.00	Pune
10009	Savings	85000.00	Hyderabad
10010	Current	110000.00	Vijayawada

Create A View For Savings Accounts:

```
CREATE VIEW Savings_Account_View AS
SELECT *
FROM Account
WHERE Account_Type = 'Savings';
SELECT * FROM Savings_Account_View;
```

Output:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

High Balance Account View:

```sql
CREATE VIEW High_Balance_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance
FROM Account
WHERE Balance > 100000;
SELECT * FROM High_Balance_View;
```

Output:

	Account_No	Customer_ID	Account_Type	Balance
▶	10003	103	Current	120000.00
	10007	107	Current	150000.00
	10010	110	Current	110000.00

Customer + Account View:

```sql
CREATE VIEW Customer_Account_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance,
    A.Branch
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID;
SELECT * FROM Customer_Account_View;
```

Output:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Sushma	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Sowmya	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Mokshagna	Bangalore	10003	Current	120000.00	Bangalore
104	Akhila	Chennai	10004	Savings	45000.00	Chennai
105	Likhitha	Hyderabad	10005	Current	90000.00	Hyderabad
106	Aswitha	Delhi	10006	Savings	65000.00	Delhi
107	Vaishnavi	Mumbai	10007	Current	150000.00	Mumbai
108	Vignesh	Pune	10008	Savings	35000.00	Pune
109	Shanmukh	Hyderabad	10009	Savings	85000.00	Hyderabad
110	Jumpa	Vijayawada	10010	Current	110000.00	Vijayawada

Complete customer bank view:

```
CREATE VIEW Customer_Banking_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance,
    A.Branch,
    L.Loan_Type,
    L.Loan_Amount
FROM Customer C
LEFT JOIN Account A
ON C.Customer_ID = A.Customer_ID
LEFT JOIN Loan L
ON C.Customer_ID = L.Customer_ID;
SELECT * FROM Customer_Banking_View;
```

Output:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch	Loan_Type	Loan_Amount
101	Sushma	Hyderabad	10001	Savings	50000.00	Hyderabad	Home Loan	5000000.00
102	Sowmya	Vijayawada	10002	Savings	75000.00	Vijayawada	Education Loan	1000000.00
103	Mokshagna	Bangalore	10003	Current	120000.00	Bangalore	Car Loan	800000.00
104	Akhila	Chennai	10004	Savings	45000.00	Chennai	Personal Loan	500000.00
105	Likhitha	Hyderabad	10005	Current	90000.00	Hyderabad	Home Loan	4000000.00
106	Aswitha	Delhi	10006	Savings	65000.00	Delhi	Car Loan	900000.00
107	Vaishnavi	Mumbai	10007	Current	150000.00	Mumbai	Business Loan	3000000.00
108	Vignesh	Pune	10008	Savings	35000.00	Pune	NULL	NULL
109	Shanmukh	Hyderabad	10009	Savings	85000.00	Hyderabad	Personal Loan	600000.00
110	Jumpa	Vijayawada	10010	Current	110000.00	Vijayawada	NULL	NULL